

P1 — Generalization as Formalized Induction

: v0.1 | : 2026-08-11 : + = (); = token
; = + : (/ /) " "
,

0.

- : *Generalization as Formalized Induction: Definition-Driven Learning in Native Token Grammars*
- ():

We propose that systematic generalization is not an emergent or a priori faculty of neural networks, but the formalization of induction: to generalize on a token A, A must be precisely defined, and every token in A’s definition chain must be sufficiently trained. Sufficient training is a relational property, not a quantitative one — it requires simultaneous training of parallel concepts under the same superordinate concept, wrong-answer training, and collocation training, so that relations between concepts become learnable signals. Training stability requires a purely native token grammar; modal separation requires that input, logic, and output grammars be designed independently, separating signifier from signified at the token level; intuition and reasoning are separated into a compiled fast path and an unfolded slow path. Sample sufficiency then follows: generalization needs no large sample volume — only precise definitions, a small set of contrastive samples, and few config-dependent epochs for the model to compile its weights (3 simple / 8 full suite).

- (=):
 1. : = , / (G0)
 2. : (G1)
 3. = : + + (G2)
 4. (G3)
 5. : / / , / token (G4)
 6. / : vs (G5)
 7. : (G6)
 - **venue:** NeurIPS/ICML/ICLR main (+); TMLR
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1.

1.0 — (G0)

G0 (): *There is no a priori or transcendental generation of conceptual ability: generalization is the formalization of induction. Any ability to generalize on a*

token must be produced — not assumed.

:

1. : “ ” ,
 - -calculus primitive (variable/abstraction/application), ” ”;
 - Church , , ;
 - ” → tokenize → Transformer” (-application/ -reduction)
- primitive
- : primitive = ” ” = ” ”
2. (): / , **token** — (E0)
3. : — (a) (: explicit/inductive/recursive/axiomatic/implicit);
- (b) token (G1-G2 ” ”)
4. : , — , ,
- : token ; + token

(E0): | # | | | | |—|—|—|—| | E0.1 | value_zero () 0 (0+n, 0×n) ; | | token skill §7.8 | | E0.2 | vocab 153 40 epochs = ; | | token skill §7.6 | | E0.3 | 0-acc token — epoch 3 1.0, 0-acc token / , epoch/ | | token skill §7.8 |

1.1 (G1)

G1 (): To generalize on token *A*, (i) *A*’s definition must be precise and complete, and (ii) every token required by *A*’s definition chain must itself be sufficiently trained. Both conditions are necessary; neither is sufficient alone.

:

1. = : A A, A ” ”
2. (i): ; / value_zero : 0 (0+n, 0×n) , (E0.1); vocab 153 40 epochs (E0.2)
3. (ii): token value_three : arrow + =1 (E1.1)
4. (i) (ii): + token (2000 / 2 / 3-8 epochs , G6; exp41 0.987, exp80) “ ” Lean (/)
5. : — () ; (, ; exp10 §7.2)
- : P1a — (value_zero): , ; P1b — token ()
- token ; P1c — (i) (ii)

(E1): | # | | | | |—|—|—|—| | E1.1 | value_three =1 () → (token) | | token skill §7.9 | | E1.2 | per_token_gen token : / token | | token skill §7.7 | | E1.3 | : value_zero 0 ; | | token skill §7.8 | | E1.4 | vocab 153 40 epochs; | | token skill §7.6 | | P1b/c | token ablation | |

(2026-08-11, exp10): 0-IMPLY (EXP-10/11,)
bug — ” 0.75/ 0.938/ ” + ; imply 0.000, =

$$+ : () ; \text{branch} \quad \text{run_exp} , \quad (\S 7.2)$$

1.2 = (G2)

G2 (): *Sufficient training is a relational property, not a quantitative one. It consists of (i) simultaneous training of parallel concepts under the same superordinate concept — making relations between concepts explicit training signals; (ii) wrong-answer training (negative examples, including structure-identical-result-wrong negatives); (iii) collocation training — every participating token’s adjacent collocations covered above a threshold.*

:
 1. : token (" ",) T7 8/9 = +
 + law + math (E2.1); all_dir_iter = × (E2.2)
 2. : " ", " "; (judgment) ()
 3. : token 3 — = ; (crash_culprit) token
 4. (exp20): (0.000), **1.0 0.833** — (De Morgan
 /inverse), (E2.8)
 5. : “ ” (/ /), ” ” — G6
 : P2a — () token (exp20); P2b — (); P2c —
 token ,

(E2): | # | | | | | | | | | | E2.1 | 8/9 (+ +law+math
) | | token skill §7.6 T7 | | E2.2 | all_dir_iter () | |
 lab/configs/all_dir_iter*.json | | E2.3 | neg 0.000→0.958: hi 9→15
 | | token skill §7.6 | | E2.4 | 3 ; crash_culprit token | | token
 skill §7.8 | | E2.5 | (exp20): iff/xor 0.798, 0.000, **1.0→0.833**;
 and/or 0.814 — , | | exp20_results.md | | P2a | (exp20);
 P2b / P2c | | |

1.3 (G3)

G3 (): *Training stability requires a purely native token grammar: the syntax itself must be made of tokens (G-layer), arranged by data-driven assembly — no externally imposed pseudo-syntax, no hardcoded arrangement knowledge.*

:
 1. : ,
 2. = (gtoken : arg/fn/args/binder/body) (P / /) (re-
 duction) , → (SKI gtoken)
 3. : token (maintain), ” - ”
 4. : ” ” / — (G4)
 : P3a — (gtoken) / ; P3b — token
 (E3): | # | | | | | | | | | | E3.1 | : assemble_seq/judge_seq
 gtoken/ptoken ; ” token” | | token skill §7.7 | | E3.2 | : / /

(check_notation) | | AGENTS.md / skill §7.7 | | E3.3 | , (EXP-61g + notation): prefix vs infix (0.836 vs 0.853, Δ 0.017; 0.677) — ; (imply) (0.969 vs 0.950); (0.948 vs 0.852) | | exp61g / notation_results.md | | E3.4 | token (EXP-61h canonical): token , canonical token (c0→1.000, c1→0.000) — (,), token () | | exp61h_results.md | | P3a/b | / ablation | | |

1.4 (G4)

G4 (): *Mirroring the structure of human cognition — perception, meaning, and expression/execution — model input, logic, and output must be assigned independent grammars. Signifier and signified must be separated at the token level (S-layer vs C-layer), so the model distinguishes concepts from their expressions from their notations from the very token layer.*

- :
1. : () → (/) → - () —
 2. : (signifier, S glyph) (signified, C) ; (glyph =/=),
 3. (P) : (/ /) , ; (fraction/radical/imaginary_unit),
numeral (digit), " value"
 4. : — () vs [] = vs () — (E4.1)
 5. : " token "
- : P4a — (/) ; P4b — (digit/operator ,) ();
P4c — /

(E4): | # | | | | | | | | | | E4.1 | () vs [] (); | |
token skill §7.8 | | E4.2 | numeral_v4_shuffle / numeral_focused_shuffle (shuffle) | | lab/configs/numeral_v4_shuffle.json | | E4.3 | numeral_v4_wrong-symmetry () | | lab/configs/numeral_v4_wrong-symmetry.json | | E4.4
| sign : sign_pos/sign_neg 3 ctoken + 4 arrow () — | | token skill
§7.8 | | E4.5 | (exp41,): 10 digit token (26080 ,), token
0.996→0.987 (0.009 0.02) — , | | exp41_results.md | | P4b |
(digit+operator) — | | |

1.5 / (G5)

G5 (, 2026-08-11): *Intuition is the reasoning fast path. It is hard to obtain — acquisition requires sufficient training (definition-complete and relation-sufficient, config-dependent epochs). Once formed, however, it generalizes easily by skipping steps: intuition jumps from input to output, bypassing the intermediate construction steps. Fast execution is its nature as a reasoning path; step-skipping generalization is its mechanism for outward generalization. Intuition is hard to obtain, but easy to generalize with.*

:

1. = (): ; , , " "

2. : — (G1) + (G2), epoch (exp53: + 8 epochs;
3) " " " — (EXP-80)

3. : — : (), ; (exp50: 2000
1.000; exp41: 0.987 —)

4. (/): (definition-driven, token_iterator) — , (=)
(): ,

5. : (fallback) — (semantic-path
compilation)

6. : () () ; " + " , epoch — G1/G6
: P5a — (); P5b — < (steps/tokens/latency); P5c —
; P5d — (EXP-80: , —)

(E5): | # | | | | | | | | | | E5.1 | (exp50): = (2/9/20/2000
) , 1.000 (2000) — | | exp50_51_results.md | | E5.2 |
token_iterator (token : digit_add +) vs () | | tok-
enizer/eval/token_iterator.py | | E5.3 | (exp51): 2000 0.62s vs 0.086s
(7×) / 0.069s (9×) | | exp50_51_results.md | | E5.4 | **vs** : exp53
(epoch 1/2/3/5/8 → 0.001/0.214/0.244/0.781/0.996, 8) + exp50/41 (2000
/) — , | | exp53_results.md | | E5.5 | (→ , ,) | |
AI .md | | E5.6 | (EXP-80): 2000 / 60 / 100% digit / ×
1.000 — (8 epochs, exp53) , SUPPORT | | exp80_results.md | |
E5.7 | (EXP-90,): 20 , 82 token/0.410ms vs 5 token/0.186ms
(num itoken, 16× token , 2.2×), = oracle : itoken vocab (pad 0),
(); " itoken", O(1) | | exp90_results.md | | P5a/b/c |
/ / | (a/b exp50/51; c) | | P5d | (EXP-80): ,
— SUPPORT | | exp80_results.md |

1.6 (G6)

G6 (): *Showing sufficiently is not showing as much as possible. Generalization requires no large sample volume — only precise definitions (G1), relational training sufficiency (G2), and few config-dependent epochs (3 simple / 8 full suite) for the model to compile its weights. The hard structural search is done by representation design; SGD only preserves the structure.*

:

1. : (× ×) — ,

2. : + , —

3. : epoch (3 / 8, exp53) = (,); epoch = /
(: num_concepts)

4. : — " " , (/ /)

: P6a — " + " , (); P6b — , 10× ; P6c —
/ token ()

(E6): | # | | | | | | | | | | E6.1 | 2000 / <1MB / 2 (,

arith_multi_type) | | docs AI .md | | E6.2 | vocab 153 40 epochs () vs
| | token skill §7.6 | | E6.3 | 0-acc token = | | token skill
§7.8 | | P6a/b/c | / / | | () |
: exp60 () , , G6 (, §3.1) exp53 (epoch)
G5” ”, E5.4

1.7 : (, **G5**)

(): — (de Bruijn/OpenMath/MathML)
(Bourbaki) , “ ” claim : () ()
() , —

R (): *A single system exhibits three execution channels for the same mathematical object — the constructive channel (unfolded definition-driven expansion), the intuitive channel (compiled weight execution), and the formal channel (definition-driven rewriting). The three channels are pairwise semantically equivalent: construction can be compiled into intuition, and both agree with formal rewriting. The three-way equivalence is the neural implementation of the Computational Trinitarianism thesis (Curry-Howard: programs = proofs = types), with equivalence holding between execution channels of one system rather than between logics.*

:	
(Brouwer) , = /	: (, ora- cle/token_iterator)
(Bourbaki) ()	: (, (,))
(Hilbert)	: (eval () engine / SKI reduce / grammar execute)

(R):

1. : ; , (,); (, /)
2. (): , — (R2)
3. O(1) (): (R1)
4. : , = = — “ , , ” Curry-Howard
, (Computational Trinitarianism)
5. (): , ; = (Brouwer) — G0

: R1 — (/ /) (Sem_ =Sem_ =Sem_); R2 —
 (); R3 — = : + epoch (, epoch 3 1.0)
 (E7): | # | | | | | | | | E7.1 | : 1-20 0.0 () |
 | .md §3 | | E7.2 | (exp50): = (2/9/20/2000), 1.000 —
 Sem_ =Sem_ =Sem_ | | exp50_51_results.md | | E7.3 | 2000
 (999...9+1): — (weights program) | | exp50_51_results.md | | E7.4 |
 (exp41): 26080 digit 0.996→0.987 — | | exp41_results.md | |
 E7.5 | epoch 3 1.0 () / 8 epochs (exp53) — | | exp53_results.md
 | | E7.6 | (EXP-80): 2000 / 60 / 100% / 1.000, 8 epochs — ,
 | | exp80_results.md | | E7.7 | = canonical (EXP-61h): token
 , token (c0→1.000, c1→0.000) — () , " = , = " |
 | exp61h_results.md |
 Lean (,): (: = ; :) + ()
 Lean — " " " " , ; Lean (mathlib claims ledger
)

2. (token)

		()
B	baseloop.jsonl (axiomatic)	G1 ; G0
C	derive.jsonl (5)	G1 ; G2
S	symbol_tokens.jsonl (glyph/maps_to)	G4 / ;
G	gtokens.jsonl (/)	G3 ; G5
P	presentation.jsonl (/ /)	G4 ;
A	arrow_tokens.jsonl (source/target/concept)	G2 ; / /

: (G5) / (G2) / run_exp () / CTE (,)

3. ()

3.1 ()

- exp02_supervised_s1/s2: 1.000 (+9 ,)
- exp10_imply_supervised: 0.996 (imply — + +)
- exp20a/b: (0.798/0.814, G2)
- exp41: (0.996→0.987, G4)

- exp50: (= = , 2000 1.000, R)
- exp51: (2000 7×, G5)
- exp53: (1/2/3/5/8 → 0.001/0.214/0.244/0.781/0.996, SUPPORT: , G5)
- exp60: (, ,)
- exp80: (2000 / 60/100% / 1.000, G5)
- exp90: (82→5 token, , G5)
- arith_multi_type / numeral_v4 / all_dir_iter / trace:

3.2

G1	token	ablation (P1b)
G2	(P2b) /	(P2c)
G3	/	(P3a/b)
G4	(P4c)	
G5	(P5c);	(exp80)
G6	()	
C1	representation	baseline ()

4. Related Work (,)

1. **-calculus** — : primitive, ; Church (Related Work,)
 2. / — Fodor (language of thought), Chomsky (): ; E0
 3. — Lake et al. 2017; Keyzers et al. 2020 (COGS); SCAN; compositionality literature: , ” ” —
 4. — Saussure (/ , ,): G4 ; =
 5. — Kahneman (System 1/2): G5 ; = ()
 6. — disentanglement/compositional structured representations: ” ”, ” ” + ”
 7. → — Learning to Compile Programs to Neural Networks (ICML 2024) : G5 , = ,
 8. (§1.7) — : (de Bruijn 1972 / OpenMath-MathML / / Bourbaki / / Mizar-Lean-Isabelle), **claim**; claim — / / , Computational Trinitarianism (Curry-Howard: programs=proofs=types) (,) : ” ”, ” : Brouwer / Bourbaki / Hilbert ; : (Martin-Löf, Lean/Coq) Gärdenfors conceptual spaces (→)
-

5. ()

- ☐ P1b (token ablation)
 - ☐ P2b/P2c (/)
 - ☐ P3a/P3b (ablation)
 - ☐ P5c () + exp80
 - ☐ G6 (3)
 - ☐ representation baseline (, C1)
 - ☐ Lean : + (/) — ,
 - ☐ (run_dir + metrics.json/views.json)
 - ☐ (× × × ×)
 - ☐ : §4 (Related Work)
 - ☐ claims ledger (status/novelty_status/)
-

(, 2026-08-11)

- , , :
- (→ : token ; attention);
 - **CoT** **flashing** (→ : ; synthetic CoT);
 - (→ : , ; distilled intuition / flashing).
- , — ASI ,